

ARTICLE

Capturing the complexity of autism: Applying a developmental cascades framework

Jessica Bradshaw¹  | Amy J. Schwichtenberg² | Jana M. Iverson³

¹University of South Carolina, Columbia, South Carolina, USA

²Purdue University, West Lafayette, Indiana, USA

³University of Pittsburgh, Pittsburgh, Pennsylvania, USA

Correspondence

Jessica Bradshaw, Department of Psychology, University of South Carolina, 1800 Gervais St., Columbia, SC 29201, USA.

Email: jbradshaw@sc.edu

Funding information

National Institute on Deafness and Other Communication Disorders, Grant/Award Number: R01DC016557 and R21DC071252; Eunice Kennedy Shriver National Institute of Child Health and Human Development, Grant/Award Number: R03HD104084; National Institute of Mental Health, Grant/Award Number: K23MH120476; James S. McDonnell Foundation

Abstract

Developmental change emerges from dynamic interactions among networks of neural activity, behavior systems, and experience-dependent processes. A developmental cascades framework captures the sequential, multilevel, cross-domain nature of human development and is ideal for demonstrating how interconnected systems have far-reaching effects in typical and atypical development. Neurodevelopmental disorders represent an intriguing application of this framework. Autism spectrum disorder (ASD) is complex and heterogeneous, with biological and behavioral features that cut across multiple developmental domains, including those that are motor, cognitive, sensory, and bioregulatory. Mapping developmental cascades in ASD can be transformational in elucidating how seemingly unrelated behaviors (e.g., those emerging at different points in development and occurring in multiple domains) are part of an interconnected neurodevelopmental pathway. In this article, we review evidence for specific developmental cascades implicated in ASD and suggest that theoretical and empirical advances in etiology and change mechanisms can be accelerated using a developmental cascades framework.

KEYWORDS

autism spectrum disorder, developmental cascades, neurodevelopment

INTRODUCTION

Developmental research has historically focused on documenting the timing and acquisition of specific skills and behaviors within established developmental domains. Infants reach by 5 months, crawl at 6 months, and walk at 12 months. Babbling emerges by 7 months and the first words can be heard by 1 year. Yet the onset of these skills varies significantly and their emergence does not occur in isolation. The onset of reaching and crawling generate new opportunities for interactions with the world. The emergence of communication quickly establishes a new motivation for locomotion. *Developmental cascades* describe how behavior and development in one

domain (e.g., motor) have cascading, far-reaching effects on other, seemingly unrelated domains (e.g., language).

The idea of developmental cascades stems from the work of key developmental scholars and theoretical models, including Gottlieb's model of developmental behavioral genetics (Gottlieb, 2003), Smith and Thelen's work on developmental dynamics (Smith & Thelen, 2003), Sameroff's transactional model of development (Sameroff, 2009), and Karmiloff-Smith's neuroconstructivist approach to neurodevelopmental disorders (NDDs; Karmiloff-Smith, 1998). A developmental cascades framework considers development as the ongoing, cumulative consequence of transactions across developmental domains, biological systems,

Abbreviations: ASD, autism spectrum disorder; CSF, cerebral spinal fluid; NDD, neurodevelopmental disorder.

© 2022 The Authors. *Child Development Perspectives* © 2022 The Society for Research in Child Development

and environmental contexts (Masten & Cicchetti, 2010). In this way, it allows researchers to examine how low-level (or high-level) features have cross-domain, far-reaching, bidirectional effects on processes later in development. Cascades enable the examination not only of specific developmental mechanisms, but also of how multiple mechanisms interact in the moment (and across time) to *shape* individual development and interactive contexts. Individual developmental trajectories diverge in response to unique experiences and events, providing countless opportunities for rich input that is driven by the individual child and consequently shapes future input.

A developmental cascades framework applied to the study of autism

Recently, a surge of interest in developmental cascades has produced several robust models for explaining cross-domain effects on overall development (e.g., Oakes & Rakison, 2019). From a developmental psychopathology perspective, these models have typically focused on older populations (e.g., Masten et al., 2005). The cascades framework provides an opportunity to extend this work to infancy, a period of rapid neurodevelopment and peak brain plasticity. In this context, such an extension may advance the discovery of etiological mechanisms and understanding of heterogenic variability in atypical developmental pathways, such as those observed in NDDs.

Autism spectrum disorder (ASD) is one such NDD, with behavioral and biological differences that persist throughout the lifespan. The promise of early behavioral interventions to support infants and toddlers with ASD has motivated mechanistic research to advance early detection. However, the heterogeneity of the disorder is vast. Sensitive and specific biological or behavioral indicators of ASD prior to 12 months remain elusive. This is largely due to our current conceptualization of ASD wherein core features (i.e., social communication differences and restricted interests/repetitive behaviors) are inherently developmental and do not fully manifest in a clearly observable way until after the first year. From a developmental cascades perspective, these core features can be conceptualized as points on a cascade that represent the cumulative result of unique input and experience, driven by the interaction between individual and contextual differences and pre- and postnatal neurological variation. Accordingly, the search for etiological mechanisms in ASD must extend beyond core features.

Etiological theories of ASD, such as the social motivation hypothesis (Chevallier et al., 2012; Dawson, 2008) elegantly describe a core impairment (i.e., social motivation) that explains the secondary (defining) characteristics of ASD. The recently developed anterior

modifiers in the emergence of neurodevelopmental disorders (AMEND) framework (Johnson et al., 2021) take a systems biology approach to understanding NDDs. A developmental cascades framework builds on these theories and allows us to understand ASD within the context of broad, cross-system, normative developmental processes, leading to rich descriptions of phenotypic heterogeneity within ASD across development. By considering the cross-domain effects of multiple experiences and behaviors that set the stage for specific developmental pathways, we can capture the complexity of neurodiversity, explain the variability and heterogeneity within ASD, and pinpoint how the timing and quality of behaviors and bioregulatory processes carry to downstream effects in development. Delineation of these developmental pathways, including biological, behavioral, and contextual contributors, can drive early detection and intervention efforts.

In some domains, infants with ASD may follow a unique developmental pathway that leads to a neurotypical outcome. For example, communicative gestures are hypothesized to be a critical point on a developmental cascade that culminates in the acquisition of first words for typically developing infants, but the use of gestures may not be part of the language acquisition cascade in ASD. Conversely, infants with ASD may show similar developmental skills early on that, because of cross-domain effects of other emerging behaviors, lead to vastly different outcomes. For example, a low-level attentional preference for high-contrast features may lead to an attentional preference for eyes in infants without ASD, but to sensory-seeking behaviors and attentional preference for objects in infants with ASD. Developmental cascades emphasize a cumulative model with additive components rather than a one-to-one mapping of single deficits that lead to maladaptive outcomes. This framework provides a unique opportunity for interventions that target a variety of domains at precise moments.

In summary, assuming that all infants experience the same developmental pathways narrows the potential for mechanism discovery. Rather than asking *when* observable features of ASD emerge, we can ask *how* they emerge. A developmental cascades framework expands our view of how cross-domain systems interact at multiple levels to influence how individual infants both experience and drive environmental input to shape their own development. Contextualizing ASD as the result of, and a contributor to, unique cascades of development that drive specific input and learning can transform and enrich our approach to research on ASD. Next, we illustrate potential developmental cascades in ASD that represent unique biobehavioral processes across three domains of development—infant sitting, visual attention, and sleep—which may be avenues for early intervention. We close with an overview of methods for using the developmental cascades framework in research on ASD and other NDDs.

DEVELOPMENTAL CASCADES

Infant sitting: A watershed moment in the first year of life

In the early months of the first year, infants spend most of their time lying supine on their backs or prone on their bellies. Between 5 and 7 months, they spend more time seated in an upright position, first with support from objects or their own hands. Over time, infants gain strength and improve their ability to balance in the sitting position, allowing them to remain upright without support for progressively longer periods.

The transition to upright, independent sitting is a watershed moment in the first year of life: It represents a significant advance in motor skill, but more importantly, it unleashes a cascade of effects in other behavioral domains (see Figure 1). This cascade provides infants with access to new experiences and opportunities for interaction with the objects and people in their environments. While infants lying supine or prone can see only the areas directly above or below the head, sitting infants have an

expanded 180-degree panoramic view of their surroundings (Luo & Franchak, 2020). Arm movements and the ability to manipulate objects are also more constrained in supine and prone positions. Supine infants must extend their arms and work against gravity to hold an object in view. In a prone position, one arm is often used for balance and support. But in the sitting position, hands are free to move and they fall naturally within infants' visual fields, making it possible to coordinate looking at objects with acting on them (e.g., Soska & Adolph, 2014), an activity that supports the extraction of rich, multimodal perceptual information.

In addition to these cascading effects, the emergence of sitting also has downstream effects on the infant-caregiver dyad. Sitting infants and their caregivers spend more time with their bodies positioned at right angles to one another, effectively enlarging the space between them and providing new opportunities for caregivers to introduce objects into shared play (Schneider et al., 2021). The construction of a large shared dyadic play space creates ideal conditions for joint attention (e.g., caregiver and infant looking at the same toy at the same time). Indeed,

Typical Motor/Sitting Cascade

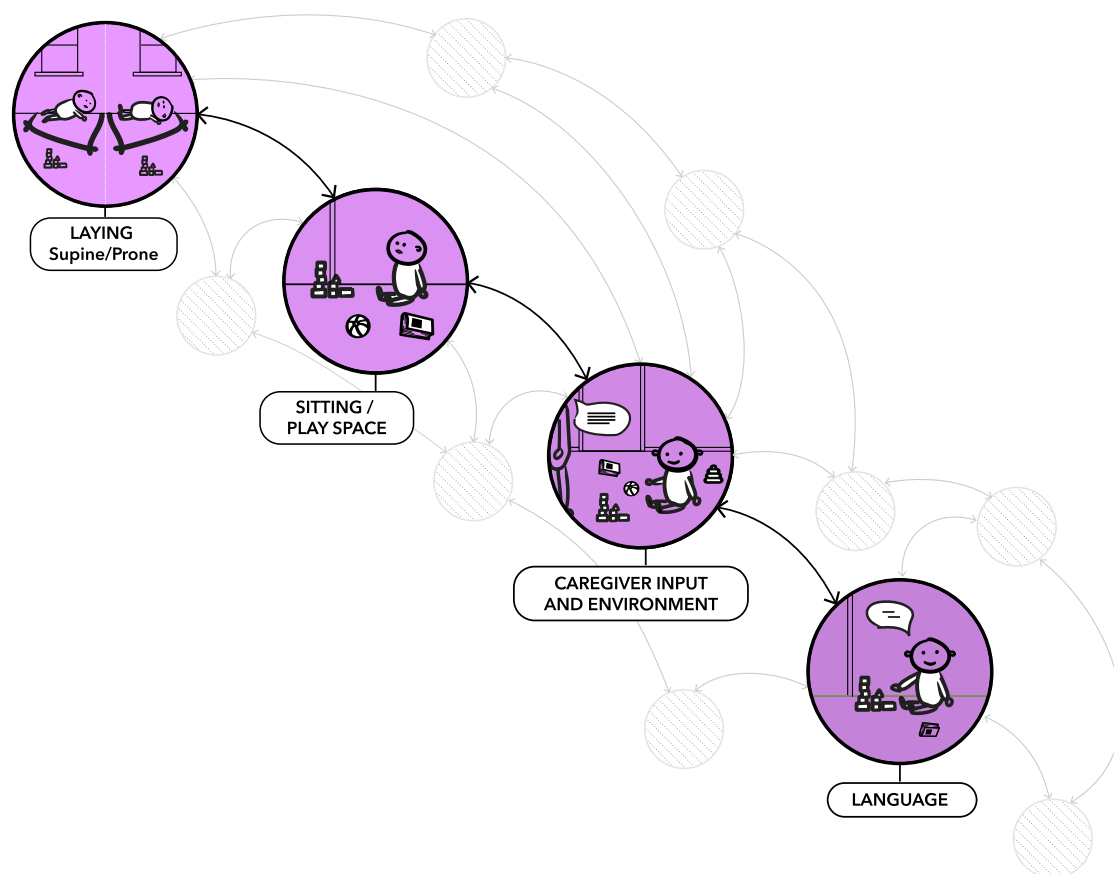


FIGURE 1 The transition from lying prone or supine to upright sitting transforms infants' experiences with objects and caregivers. Infants' ability to coordinate reaching and grasping objects with looking at a caregiver provides new opportunities for rich language input from the caregiver

dyads spend significantly more time in joint attention when infants are positioned in sitting compared to prone positions (Franchak et al., 2018). In addition, when caregivers provide rich language input about an object of interest, they create powerful opportunities for language learning.

Developmental differences in gross and fine motor skills are becoming a recognized feature of emerging ASD (e.g., Licari et al., 2021). Infants with ASD experience atypical head control compared to infants without ASD (Flanagan et al., 2012). On average, infants with ASD begin to sit independently later than their neurotypically developing peers (Nickel et al., 2013). In addition, even when they are able to sit independently, infants with ASD spend significantly more time in lying postures (prone, supine) than infants without ASD, likely because the sitting posture requires more protracted development of balance and control (Leezenbaum & Iverson, 2019). Less time spent sitting upright and slowed consolidation of sitting skills constrain opportunities for more sophisticated object manipulation and play. Moreover, because caregivers use infants' manual actions as a pathway to joint attention (Yu & Smith, 2017) and are more likely to label objects their infant is holding, looking at, and actively manipulating (West & Iverson, 2017), less object manipulation in infants with ASD may translate into reduced opportunities for joint attention and alterations in caregivers' language input. Thus, while the development of sitting may appear to be unrelated to the development of social communication and language, evidence suggests that sitting has cascading developmental effects on behaviors within and beyond the motor domain, and within and beyond the infant. For infants with ASD, differences in the timing and consolidation of sitting and its integration with other behaviors (e.g., object manipulation) may affect the unfolding of the developmental cascade in ways that fundamentally alter their experiences with objects and people.

Visual attention: A complex skill with cascading effects

Endogenous attention develops rapidly over the first months of life. The emergence of volitional, selective attention by 6–10 weeks is subserved by a transition from subcortical to cortical control of behavior (Johnson et al., 1991). Efficient disengagement from salient stimuli, attention shifting, and filtering of extraneous information are all critical components of selective attention (Atkinson et al., 1992). Developing attention in infancy is both a product of and a contributor to multiple cross-domain developmental cascades (see Figure 2). As described earlier, motoric achievements create new access to visual input. The coupling of motor and sensory systems, including coordination of eyes, head, hands, and body, is inherently linked to selective attention. Attention is

affected by cognition and experience, and cognition is shaped in part by attention. The ubiquitous role of attention, sensorimotor coupling, and experience on higher- and lower-order processes throughout development makes it an intriguing access point for understanding etiological mechanisms and unique developmental pathways in ASD.

Atypical visual attention patterns, particularly to socially relevant features of the environment, constitute an early-emerging and enduring feature of ASD across the lifespan (Frazier et al., 2017). Many mechanistic theories of autism incorporate attention differences as a core feature of the ASD phenotype: social motivation (decreased social attention; Chevallier et al., 2012), central coherence (global vs. local visual processing; Happé & Frith, 2006), executive function (attention disengagement; Pellicano, 2012). Differences in lower-level attention in ASD begin very early in development, evidenced by decreased orienting to and tracking of faces and objects (Bradshaw et al., 2020, 2021), decreasing attention to the eyes and face (Jones & Klin, 2013), and difficulties with attentional disengagement and shifting (Elsabbagh et al., 2013). Atypical attention is not in itself a core diagnostic feature of ASD, but instead represents a subtle, early-emerging variation that has proximal and distal effects on individuals' perception of and interaction with the environment.

For example, eyes are visually salient and newborns demonstrate an attentional preference for eyes from the first hours of life. At a higher level, eyes are laden with rich social and communicative information, suggesting that decreased attentional preference for eyes in infancy can result in missed critical social learning opportunities. Moreover, because familiarity is an important component of selective attention, an early reduction in eye-looking may lead to increasingly diminished eye-looking over time, cumulatively resulting in reduced social attention and a nonsocial attentional preference (e.g., Pierce et al., 2016).

What causes this early variability in such low-level attentional behaviors? Diminished attention disengagement has been associated with atypical neural circuitry and suggested as a domain-general starting point for a developmental cascade that leads to social-specific deficits in ASD (Baranek et al., 2018). Recent work suggests atypical neural and neurochemical circuitry specific to visual processing and arousal modulation, which may constitute an even earlier point in a developmental cascade related to attention differences in ASD (Artoni et al., 2020; Hazlett et al., 2017).

Together, this research points to a potential cascade that crosses biology and behavior and begins shortly after birth. Differences in neurotransmitter levels and neural connectivity, particularly as they relate to arousal, attention, and sensory systems, may lead to behavioral variability in lower-order attentional processes, including attention orienting, disengagement, and shifting.

Typical Attention Cascade

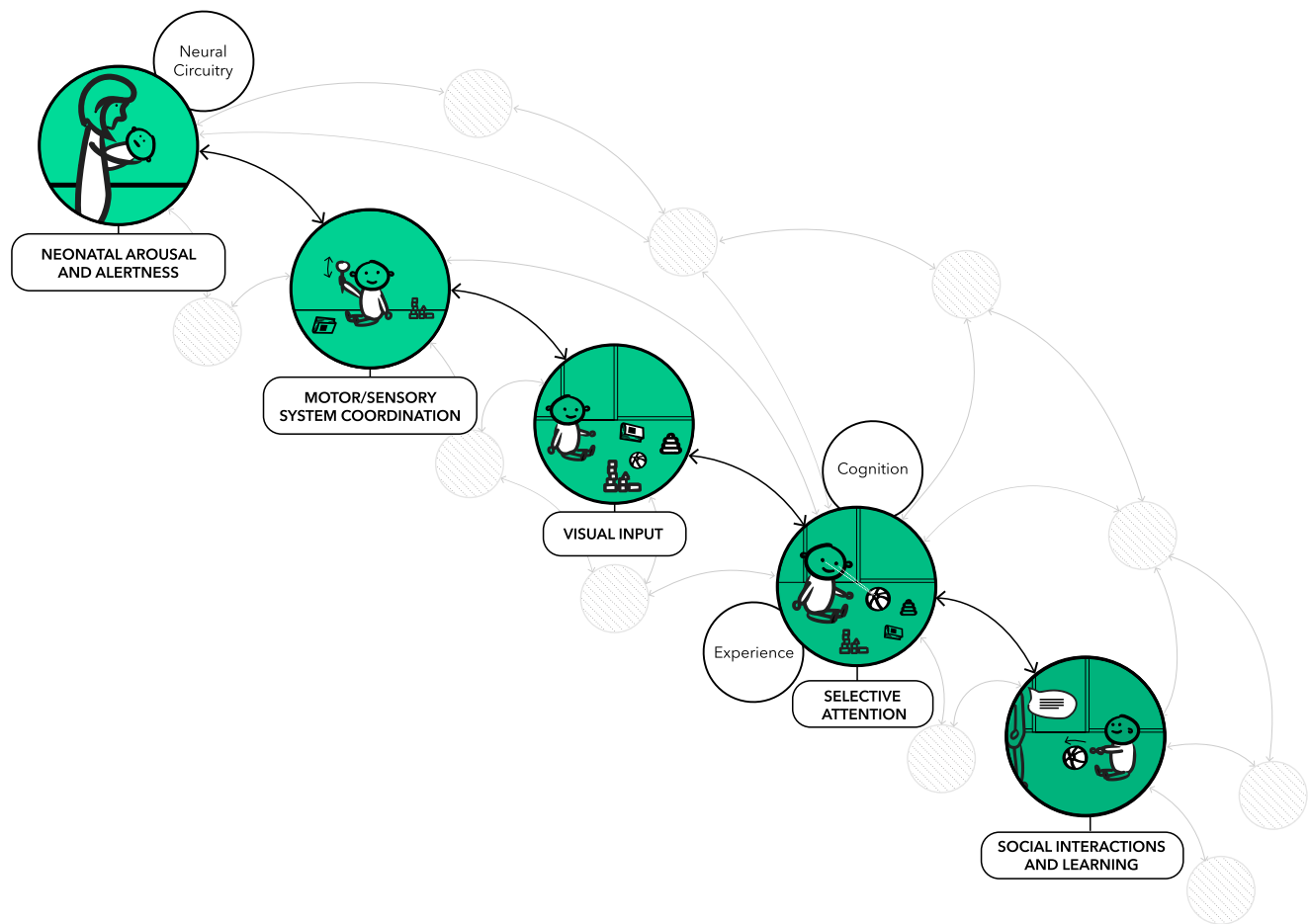


FIGURE 2 The development of attention begins in the neonatal period with modulated levels of arousal and periods of quiet alertness. Subsequent motor, sensory, and cognitive development allow for triadic (infant-object-caregiver) interactions, which provide increased opportunities for social interaction and language

These early, subtle attentional differences can have cascading effects on social interactions and learning. A reduced capacity for attention shifting in early infancy may decrease opportunities for triadic (caregiver-object-infant) interactions. This may interfere with integrating higher-order cognitive concepts, which requires jointly attending to both caregiver and object, and combining objects together in play—skills that are critical for language development. Infants drive caregiver-provided opportunities for learning and social interaction and, in ASD, individual differences in attention and the neural structures that support it may have cascading effects on social experience and learning.

Sleep: A critical bidirectional, bioregulatory behavior in infancy

Sleep is one of the first bioregulatory behaviors infants must master to support neurologic, digestive, and immune health. Infants with consistent sleep problems

early in life are at elevated risk for problems across several developmental domains, including attention, learning, memory, language, elevated adiposity, emotion regulation, and social skills. Sleep problems are well-documented across life for individuals with ASD (Cohen et al., 2014) but the potential and affiliated developmental cascades are understudied. For example, the synaptic homeostasis hypothesis of sleep (Tononi & Cirelli, 2014) posits that a core function of sleep is to selectively prune synaptic connections to reinforce those used during the day, thus making established pathways more efficient. For children with ASD, applying a developmental cascade that starts with early sleep deficiency could help explain the mechanistic link between ASD and synaptic overgrowth (Courchesne et al., 2003), large brain volume and head size (Hazlett et al., 2017), and reduced synaptic efficiency (Just et al., 2012).

For example, excess extra-axial cerebral spinal fluid (CSF) has been documented in three groups of infants who later received a diagnosis of ASD (Shen, 2018). This could reflect a cascade that starts with sleep dysfunction.

Circulation of CSF in the brain occurs predominately during sleep, specifically during slow-wave sleep or non-rapid eye movement stage 3 sleep (Fultz et al., 2019). Disrupted sleep may compromise efficient CSF circulation, possibly resulting in the atypical CSF disbursement observed in ASD. Indeed, recent studies of ASD in early infancy show altered sleep patterns and direct links between early sleep problems and atypical brain development (MacDuffie et al., 2020).

A developmental cascades approach to the connection between sleep and behavior in ASD could also inform our mechanistic understanding. Deficient sleep in ASD is associated with the severity of core symptoms as well as co-occurring features (e.g., challenging behaviors, attention problems), with notable links to arousal and repetitive behaviors (Cohen et al., 2014). Sleep deprivation affects arousal modulation profoundly. This association is characterized by an initial increase in arousal with the onset of sleep deprivation, followed by a chronic state of underarousal with extended periods of deprivation (Tobaldini et al., 2017). In ASD, underarousal is associated with sleep dysregulation and challenging behaviors (Cohen et al., 2011).

The field's current practice of assessing behavior within domains (e.g., types of sleep problems, repetitive behavior profiles) can lead to missed opportunities to recognize developmental mechanistic pathways of influence. Applying a developmental cascades framework

explicitly draws us toward cross-domain mechanisms and helps us understand how these relations change and grow across development (see Figure 3). In this example, a chronic sleep deficit that emerges in infancy may feed into a dampened arousal system. Consequently, in an attempt to modulate arousal, individuals with ASD may exhibit elevated repetitive or challenging behaviors. Therefore, repetitive behaviors may be one behavioral manifestation of underlying disrupted arousal modulation, a bioregulatory process that is inherently linked to atypical sleep and brain development.

RESEARCH METHODS AND ANALYTIC MODELS

In developmental science, most agree that applying theory is a best practice when establishing questions of interest or testable models. However, analytic models (and their limitations) often drive the types of questions posed in research and how they are tested. To help in the application of a developmental cascades framework in ASD research, in the next section, we outline several analytic approaches that marry well with this framework. The list is not exhaustive, prescriptive, or specific to the cascades framework. Rather, it exemplifies how developmental cascades and analytic models can be symbiotic.

Atypical Sleep Cascade

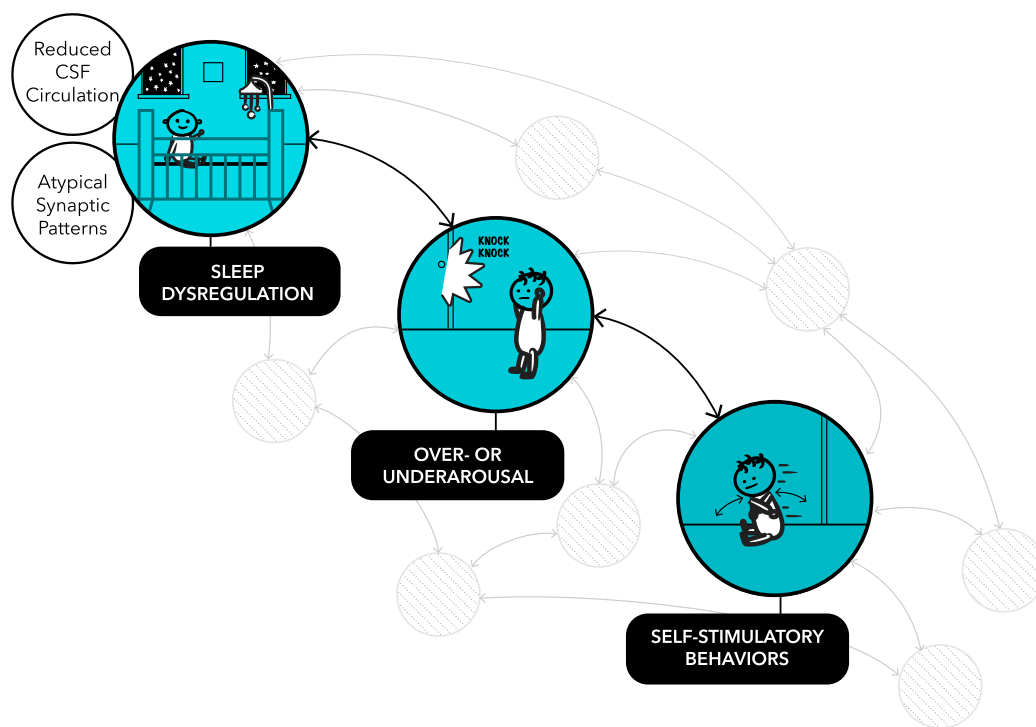


FIGURE 3 Atypical sleep early in life can influence several elements of brain development (e.g., synaptic pattern formation and pruning, cerebrospinal fluid [CSF] circulation) and biosocial processes (e.g., arousal). Over- or underarousal, in turn, is associated with self-stimulatory behavior as individuals aim to modulate their arousal using behavioral/physical actions

The initial application of a theoretical approach often includes *mapping* or specifying key factors or the pathway of interest. The accessible cause-outcome representation and notation system (Moore & George, 2011) is a tool for visualizing and modeling developmental pathways, including bidirectional effects, within- and cross-domain developmental change, and specifications for variance patterns and relative developmental timing. When testing a developmental cascade, researchers often use longitudinal designs and the most common analytic strategy is path model testing, which allows for simultaneously estimating several paths or associations with possible adjustments for model complexity, nesting, and auto-correlated features (e.g., Lynne-Landsman et al., 2010; Masten et al., 2005). For example, in one testable pathway model, sleep problems affected emotion-regulation difficulties persistently, which in turn affected attention regulation (Williams et al., 2015). Not only did this model support a sleep-emotion-attention cascade, but it also revealed transactional processes, providing a clear example of how one analytic approach can be used to combine theoretical frameworks.

When several cascades are possible, a comparative model approach can assess which model accounts for most of the variability in an outcome of interest. A supplemental approach that is sometimes used is a likelihood assessment to index the likelihood of a target outcome (e.g., ASD diagnosis) with the presence, absence, or varying degrees of other independent variables. In one example, researchers assessed the percentage of children classified as being at elevated risk for a particular outcome (in this case, substance abuse) at each stage of the proposed cascade (Dodge et al., 2009). Ultimately, few children followed the entire specified cascade, but inflection points of risk were identified and later used to inform interventions. Finally, researchers have used regression-based models of mediation and moderation, but this approach is less common and typically tests relatively short cascades.

CONCLUSION

The search for etiological mechanisms of ASD has proven to be a challenge, largely driven by the heterogeneous, cross-domain, developmental nature of the disorder. Core features of ASD change with development and manifest differently as a result of unique individual experiences and skills. In other words, the ASD phenotype is rich with variation across biological and behavioral domains. Studies of infants with ASD have expanded our understanding of the phenotype and highlighted nuanced, subtle variations in domains that do not necessarily constitute core features of the disorder. Novel discoveries can be made and the field advanced when the complexity of ASD is harnessed, rather than controlled for or reduced.

In this article, we described three developmental cascades that can help transform our understanding of ASD and social development. The development of postural control and the emergence of sitting influences attention, play, and communicative opportunities profoundly. Subtle differences in visual attention and underlying neurological systems can have cascading effects on sensorimotor behavior, social attention, and communication. Sleep is one of the most important bioregulatory processes that supports arousal modulation and may underlie some of the most challenging behaviors associated with ASD. Pathways within these cascades can occur bidirectionally in which higher-level systems (e.g., play skills) can affect lower-level systems (e.g., muscle development), and while not all domains at all levels can be measured in one study, this framework can be used to develop new studies that examine new mechanisms. These examples are intended to catalyze the use of developmental cascades in how we conceptualize studies, analyze data, and interpret findings. While the cascades we have outlined were inspired by potential areas of disruption in ASD, they are not specific to ASD. A developmental cascades framework can inform the sequential, multilevel, cross-domain nature of autism and demonstrate how interconnected systems have far-reaching effects in typical and atypical development across the lifespan.

ACKNOWLEDGMENTS

Work on this article was supported by grants from the National Institute of Mental Health (K23 MH120476 to Jessica Bradshaw), the National Institute of Deafness and Communication Disorders (R21 DC071252 to Jessica Bradshaw and R01DC016557 to Jana Iverson), and the Eunice Kennedy Shriver National Institute of Child Health and Human Development (R03 HD104084 to Amy J. Schwichtenberg).

ORCID

Jessica Bradshaw  <https://orcid.org/0000-0003-3003-3738>

REFERENCES

- Arttoni, P., Piffer, A., Vinci, V., LeBlanc, J., Nelson, C. A., Hensch, T. K., & Fagioli, M. (2020). Deep learning of spontaneous arousal fluctuations detects early cholinergic defects across neurodevelopmental mouse models and patients. *Proceedings of the National Academy of Sciences of the United States of America*, 117(38), 23298–23303. <https://doi.org/10.1073/PNAS.1820847116>
- Atkinson, J., Hood, B., Wattam-Bell, J., & Braddick, O. (1992). Changes in infants' ability to switch visual attention in the first three months of life. *Perception*, 21(5), 643–653. <https://doi.org/10.1068/p210643>
- Baranek, G. T., Woynarowski, T. G., Nowell, S., Turner-Brown, L., DuBay, M., Crais, E. R., & Watson, L. R. (2018). Cascading effects of attention disengagement and sensory seeking on social symptoms in a community sample of infants at-risk for a future diagnosis of autism spectrum disorder. *Developmental Cognitive Neuroscience*, 29(January), 30–40. <https://doi.org/10.1016/j.dcn.2017.08.006>
- Bradshaw, J., Klin, A., Evans, L., Klaiman, C., Saulnier, C., & McCracken, C. (2020). Development of attention from birth

- to 5 months in infants at risk for autism spectrum disorder. *Development and Psychopathology*, 32(2), 491–501. <https://doi.org/10.1017/S0954579419000233>
- Bradshaw, J., Shi, D., Hendrix, C. L., Saulnier, C., & Klaiman, C. (2021). Neonatal neurobehavior in infants with autism spectrum disorder. *Developmental Medicine & Child Neurology*. <https://doi.org/10.1111/dmcn.15096>
- Chevallier, C., Kohls, G., Troiani, V., Brodtkin, E. S., & Schultz, R. T. (2012). The social motivation theory of autism. *Trends in Cognitive Sciences*, 16(4), 231–239. <https://doi.org/10.1016/J.TICS.2012.02.007>
- Cohen, I. L., Yoo, J. H., Goodwin, M. S., & Moskowitz, L. (2011). Assessing challenging behaviors in autism spectrum disorders: Prevalence, rating scales, and autonomic indicators. In J. Matson & P. Sutmev (Eds.), *International handbook of autism and pervasive developmental disorders*. *Autism and Child Psychopathologies Series* (pp. 247–270). Springer. https://doi.org/10.1007/978-1-4419-8065-6_15
- Cohen, S., Conduit, R., Lockley, S. W., Rajaratnam, S. M., & Cornish, K. M. (2014). The relationship between sleep and behavior in autism spectrum disorder (ASD): A review. *Journal of Neurodevelopmental Disorders*, 6(1), 1–10. <https://doi.org/10.1186/1866-1955-6-44>
- Courchesne, E., Carper, R., & Akshoomoff, N. (2003). Evidence of brain overgrowth in the first year of life in autism. *Journal of the American Medical Association*, 290(3), 337–344. <https://doi.org/10.1001/JAMA.290.3.337>
- Dawson, G. (2008). Early behavioral intervention, brain plasticity, and the prevention of autism spectrum disorder. *Development and Psychopathology*, 20(03), 775–803. <https://doi.org/10.1017/S0954579408000370>
- Dodge, K. A., Malone, P. S., Lansford, J. E., Miller, S., Pettit, G. S., & Bates, J. E. (2009). A dynamic cascade model of the development of substance-use onset. *Monographs of the Society for Research in Child Development*, 74(3), vii–119. <https://doi.org/10.1111/J.1540-5834.2009.00528.X>
- Elsabbagh, M., Fernandes, J., Jane Webb, S., Dawson, G., Charman, T., Johnson, M. H.; British Autism Study of Infant Siblings Team. (2013). Disengagement of visual attention in infancy is associated with emerging autism in toddlerhood. *Biological Psychiatry*, 74(3), 189–194. <https://doi.org/10.1016/j.biopsych.2012.11.030>
- Flanagan, J. E., Landa, R., Bhat, A., & Bauman, M. (2012). Head lag in infants at risk for autism: A preliminary study. *American Journal of Occupational Therapy*, 66(5), 577–585. <https://doi.org/10.5014/ajot.2012.004192>
- Franchak, J. M., Kretch, K. S., & Adolph, K. E. (2018). See and be seen: Infant-caregiver social looking during locomotor free play. *Developmental Science*, 21(4), e12626. <https://doi.org/10.1111/desc.12626>
- Frazier, T. W., Strauss, M., Klingemier, E., Zetzer, E., Hardan, A. Y., Eng, C., & Youngstrom, E. A. (2017). A meta-analysis of gaze differences to social and nonsocial information between individuals with and without autism. *Journal of the American Academy of Child and Adolescent Psychiatry*, 56(7), 546–555. <https://doi.org/10.1016/J.JAAC.2017.05.005>
- Fultz, N. E., Bonmassar, G., Setsompop, K., Stickgold, R. A., Rosen, B. R., Polimeni, J. R., & Lewis, L. D. (2019). Coupled electrophysiological, hemodynamic, and cerebrospinal fluid oscillations in human sleep. *Science*, 366(6465), 628–631. <https://doi.org/10.1126/science.aax5440>
- Gottlieb, G. (2003). On making behavioral genetics truly developmental. *Human Development*, 46(6), 337–355. <https://doi.org/10.1159/000073306>
- Happé, F., & Frith, U. (2006). The weak coherence account: Detail-focused cognitive style in autism spectrum disorders. *Journal of Autism and Developmental Disorders*, 36(1), 5–25. <https://doi.org/10.1007/S10803-005-0039-0>
- Hazlett, H. C., Gu, H., Munsell, B. C., Kim, S. H., Styner, M., Wolff, J. J., Elison, J. T., Swanson, M. R., Zhu, H., Botteron, K. N., Collins, D. L., Constantino, J. N., Dager, S. R., Estes, A. M., Evans, A. C., Fonov, V. S., Gerig, G., Kostopoulos, P., McKinstry, R. C., ... Piven, J. (2017). Early brain development in infants at high risk for autism spectrum disorder. *Nature*, 542(7641), 348–351. <https://doi.org/10.1038/nature21369>
- Johnson, M. H., Charman, T., Pickles, A., & Jones, E. J. H. (2021). Annual research review: Anterior Modifiers in the Emergence of Neurodevelopmental Disorders (AMEND)—A systems neuroscience approach to common developmental disorders. *Journal of Child Psychology and Psychiatry*, 62(5), 610–630. <https://doi.org/10.1111/jcpp.13372>
- Johnson, M., Dziurawiec, S., Ellis, H., & Morton, J. (1991). Newborns' preferential tracking of face-like stimuli and its subsequent decline. *Cognition*, 40(1–2), 1–19. [https://doi.org/10.1016/0010-0277\(91\)90045-6](https://doi.org/10.1016/0010-0277(91)90045-6)
- Jones, W., & Klin, A. (2013). Attention to eyes is present but in decline in 2–6-month-old infants later diagnosed with autism. *Nature*, 504(7480), 427–431. <https://doi.org/10.1038/nature12715>
- Just, M. A., Keller, T. A., Malave, V. L., Kana, R. K., & Varma, S. (2012). Autism as a neural systems disorder: A theory of frontal-posterior underconnectivity. *Neuroscience & Biobehavioral Reviews*, 36(4), 1292–1313. <https://doi.org/10.1016/j.neubiorev.2012.02.007>
- Karmiloff-Smith, A. (1998). Development itself is the key to understanding developmental disorders. *Trends in Cognitive Sciences*, 2(10), 389–398. [https://doi.org/10.1016/S1364-6613\(98\)01230-3](https://doi.org/10.1016/S1364-6613(98)01230-3)
- Leezenbaum, N. B., & Iverson, J. M. (2019). Trajectories of posture development in infants with and without familial risk for autism spectrum disorder. *Journal of Autism and Developmental Disorders*, 49(8), 3257–3277. <https://doi.org/10.1007/s10803-019-04048-3>
- Licari, M. K., Varcin, K., Hudry, K., Leonard, H. C., Alvares, G. A., Pillar, S. V., Stevenson, P. G., Cooper, M. N., Whitehouse, A. J. O.; the AICES Team. (2021). The course and prognostic capability of motor difficulties in infants showing early signs of autism. *Autism Research*, 14(8), 1759–1768. <https://doi.org/10.1002/aur.2545>
- Luo, C., & Franchak, J. M. (2020). Head and body structure infants' visual experiences during mobile, naturalistic play. *PLoS One*, 15(11), e0242009. <https://doi.org/10.1371/JOURN.AL.PONE.0242009>
- Lynne-Landsman, S. D., Bradshaw, C. P., & Ialongo, N. S. (2010). Testing a developmental cascade model of adolescent substance use trajectories and young adult adjustment. *Development and Psychopathology*, 22(4), 933–948. <https://doi.org/10.1017/S0954579410000556>
- MacDuffie, K. E., Shen, M. D., Dager, S. R., Styner, M. A., Kim, S. H., Paterson, S., Pandey, J., St. John, T., Elison, J. T., Wolff, J. J., Swanson, M. R., Botteron, K. N., Zwaigenbaum, L., Piven, J., & Estes, A. M. (2020). Sleep onset problems and subcortical development in infants later diagnosed with autism spectrum disorder. *American Journal of Psychiatry*, 177(6), 518–525. <https://doi.org/10.1176/appi.ajp.2019.19060666>
- Masten, A. S., & Cicchetti, D. (2010). Developmental cascades. *Development and Psychopathology*, 22(3), 491–495. <https://doi.org/10.1017/S0954579410000222>
- Masten, A. S., Roisman, G. I., Long, J. D., Burt, K. B., Obradović, J., Riley, J. R., Boelcke-Stennes, K., & Tellegen, A. (2005). Developmental cascades: Linking academic achievement and externalizing and internalizing symptoms over 20 years. *Developmental Psychology*, 41(5), 733–746. <https://doi.org/10.1037/0012-1649.41.5.733>
- Moore, D. G., & George, R. (2011). ACORNS: A tool for the visualisation and modelling of atypical development. *Journal of Intellectual Disability Research*, 55(10), 956–972. <https://doi.org/10.1111/J.1365-2788.2011.01471.X>
- Nickel, L. R., Thatcher, A. R., Keller, F., Wozniak, R. H., & Iverson, J. M. (2013). Posture development in infants at heightened versus low risk for autism spectrum disorders. *Infancy*, 18(5), 639–661. <https://doi.org/10.1111/inf.12025>

- Oakes, L., & Rakison, D. (2019). *Developmental cascades: Building the infant mind*. Oxford University Press. <https://doi.org/10.1093/oso/9780195391893.003.0005>
- Pellicano, E. (2012). The development of executive function in autism. *Autism Research and Treatment*, 2012(146132), 1–8. <https://doi.org/10.1155/2012/146132>
- Pierce, K., Marinero, S., Hazin, R., McKenna, B., Barnes, C. C., & Malige, A. (2016). Eye tracking reveals abnormal visual preference for geometric images as an early biomarker of an autism spectrum disorder subtype associated with increased symptom severity. *Biological Psychiatry*, 79(8), 657–666. <https://doi.org/10.1016/j.biopsych.2015.03.032>
- Sameroff, A. (2009). The transactional model. In A. Sameroff (Ed.), *The transactional model of development: How children and contexts shape each other* (pp. 3–21). American Psychological Association. <https://doi.org/10.1037/11877-001>
- Schneider, J. L., Roemer, E. J., Northrup, J. B., & Iverson, J. M. (2021). *Dynamics of the dyad: How mothers and infants co-construct interaction spaces during object play* [Manuscript in preparation]. Department of Psychology, University of Pittsburgh.
- Shen, M. D. (2018). Cerebrospinal fluid and the early brain development of autism. *Journal of Neurodevelopmental Disorders*, 10(39), 1–10. <https://doi.org/10.1186/S11869-018-9256-7>
- Smith, L. B., & Thelen, E. (2003). Development as a dynamic system. *Trends in Cognitive Sciences*, 7(8), 343–348. [https://doi.org/10.1016/S1364-6613\(03\)00156-6](https://doi.org/10.1016/S1364-6613(03)00156-6)
- Soska, K. C., & Adolph, K. E. (2014). Postural position constrains multimodal object exploration in infants. *Infancy*, 19(2), 138–161. <https://doi.org/10.1111/INFA.12039>
- Tobaldini, E., Costantino, G., Solbiati, M., Cogliati, C., Kara, T., Nobili, L., & Montano, N. (2017). Sleep, sleep deprivation, autonomic nervous system and cardiovascular diseases. *Neuroscience & Biobehavioral Reviews*, 74(March), 321–329. <https://doi.org/10.1016/J.NEUBIOREV.2016.07.004>
- Tononi, G., & Cirelli, C. (2014). Sleep and the price of plasticity: From synaptic and cellular homeostasis to memory consolidation and integration. *Neuron*, 81(1), 12–34. <https://doi.org/10.1016/J.NEURON.2013.12.025>
- West, K. L., & Iverson, J. M. (2017). Language learning is hands-on: Exploring links between infants' object manipulation and verbal input. *Cognitive Development*, 43(July), 190–200. <https://doi.org/10.1016/J.COGEDEV.2017.05.004>
- Williams, K. E., Berthelsen, D., Walker, S., & Nicholson, J. M. (2015). A developmental cascade model of behavioral sleep problems and emotional and attentional self-regulation across early childhood. *Behavioral Sleep Medicine*, 15(1), 1–21. <https://doi.org/10.1080/15402002.2015.1065410>
- Yu, C., & Smith, L. B. (2017). Multiple sensory-motor pathways lead to coordinated visual attention. *Cognitive Science*, 41(Suppl 1), 5–31. <https://doi.org/10.1111/cogs.12366>

How to cite this article: Bradshaw, J., Schwichtenberg, A. J., & Iverson, J. M. (2022). Capturing the complexity of autism: Applying a developmental cascades framework. *Child Development Perspectives*, 16, 18–26. <https://doi.org/10.1111/cdep.12439>